Lecture 28
Comparing two means

Review

1 What is the pooled proportion?

$\hat{p} = (X_1 + X_2)/(n_1 + n_2)$: all the successes over all the observations in both samples

2 What are independent samples?

one sample tells us nothing about the other: separate random samples, or randomized groups

3 What are dependent samples?

each value is paired with one in the other sample: the same subjects twice, twins

Review: to pool or not? (two proportions)

null hypothesis or goal	the SE we use	pool?
$H_0: p_1 = p_2$	$\text{SE}_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}, \quad \hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$	**pooled**
$H_0: p_1 - p_2 = d_0,\ d_0 \neq 0$	$\text{SE} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	**unpooled**
a $100(1-\alpha)\%$ CI for $p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	**unpooled**

pool only when H_0 says $p_1 = p_2$: then there is one p, and both samples estimate it

Two means: the notation

populations 1 and 2

the two groups we compare (for example, girls and boys)

$\mu_1,\ \mu_2$

the population means: the parameters

$\sigma_1,\ \sigma_2$

the population standard deviations: almost always unknown

$n_1, \overline{x}_1, s_1;\ \ n_2, \overline{x}_2, s_2$

the size, mean and standard deviation of each sample

$\overline{x}_1 - \overline{x}_2$

the estimate: $\overline{x}_1 - \overline{x}_2$ estimates $\mu_1 - \mu_2$

Two means: the hypotheses

$H_0: \mu_1 = \mu_2$

the same as $H_0: \mu_1 - \mu_2 = 0$: no difference between the populations

$H_A: \mu_1 - \mu_2 > 0$

upper-tailed: population 1 has the larger mean

$H_A: \mu_1 - \mu_2 < 0$

lower-tailed: population 1 has the smaller mean

$H_A: \mu_1 - \mu_2 \neq 0$

two-sided: a difference in either direction

The sampling distribution of $\bar{x}_1 - \bar{x}_2$

its center

$\mu_1 - \mu_2$: **on average, the estimate $\overline{x}_1 - \overline{x}_2$ hits the true difference**

its SE: variances add for independent samples

$$\text{SE} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

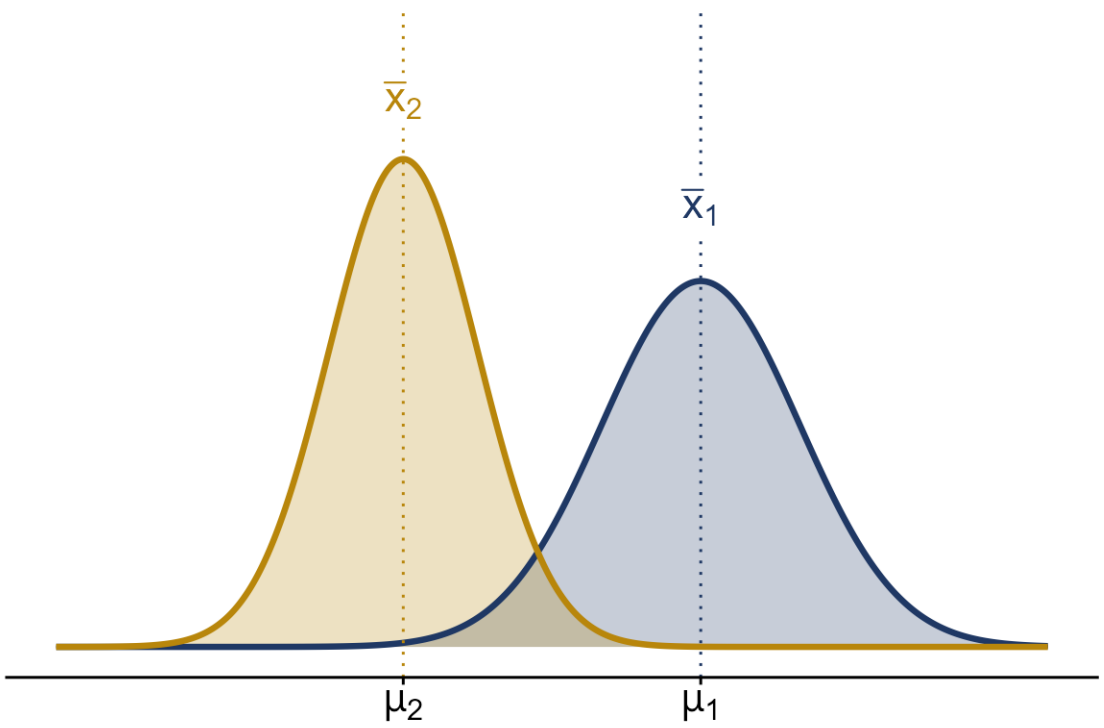
its shape

normal if both populations are normal; approximately normal if both $n \geq 30$

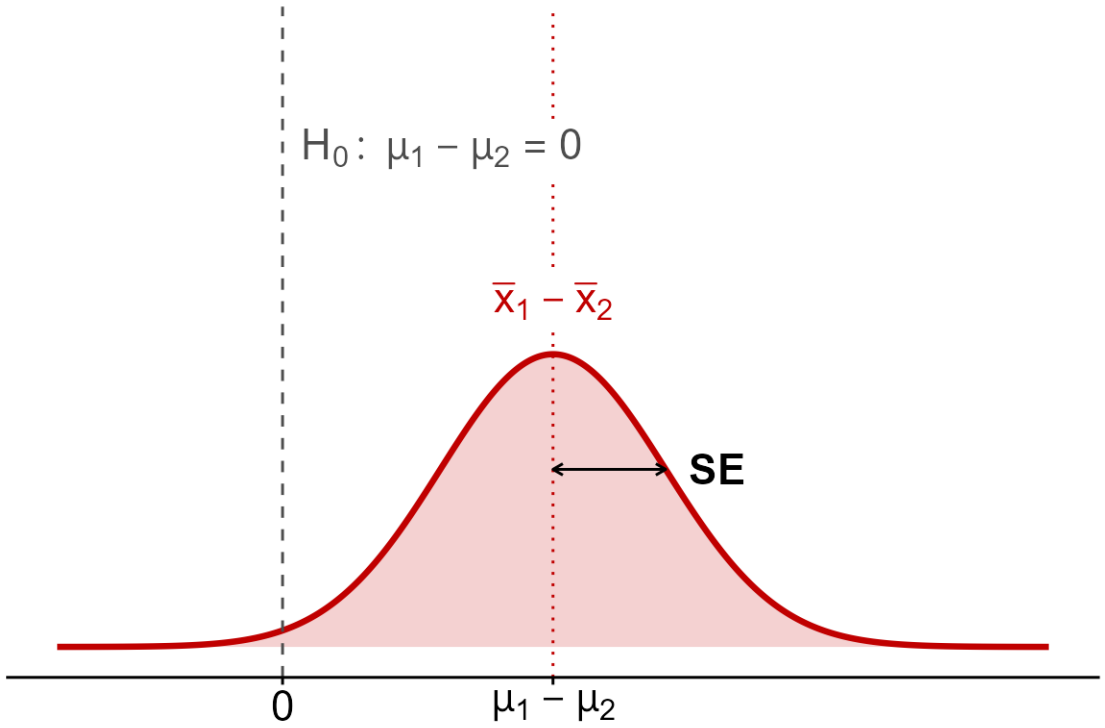
its distribution: approximately, once s_1, s_2 replace σ_1, σ_2 in the SE

$$\frac{(\overline{x}_1 - \overline{x}_2) - (\mu_1 - \mu_2)}{\text{SE}} \sim t(\min(n_1 - 1, n_2 - 1))$$

Picturing $\bar{x}_1 - \bar{x}_2$



two sample means, each with its own SE



the difference: wider, because variances add

$\bar{x}_1 - \bar{x}_2$ is centred at $\mu_1 - \mu_2$, and more spread out than either mean: the variances add

The two independent samples t-test

σ_1, σ_2 unknown

use s_1, s_2: $\mathrm{SE} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$

the test statistic

$$t_{obs} = \frac{(\overline{x}_1 - \overline{x}_2) - 0}{\mathrm{SE}}$$

degrees of freedom

$df \approx \min(n_1 - 1, n_2 - 1)$: **an approximation (next slide)**

the P-value

the t-table with that df: the tail(s) H_A points to

assumptions

two independent random samples; both populations normal, or both $n \geq 30$

Degrees of freedom: an approximation

the Welch df (what software uses), with $v_1 = s_1^2/n_1,\ v_2 = s_2^2/n_2$

$$df = \frac{(v_1 + v_2)^2}{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)}$$

in practice, by hand

$df \approx \min(n_1 - 1, n_2 - 1)$: **a quick approximation to the Welch df**

why that is safe

it is never larger than the Welch df, so our P-values come out a little too big: conservative

either way, an approximation

t_{obs} **only approximately follows a t distribution, even with the Welch df**

Two means: to pool or not?

test or interval	the SE (and df) we use	pool?
t-test, assume $\sigma_1 = \sigma_2$	$\text{SE} = s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}, \quad df = n_1 + n_2 - 2$	**pooled**
t-test, σ's may differ	$\text{SE} = \sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}, \quad df \approx \min(n_1 - 1, n_2 - 1)$	**unpooled**
$100(1-\alpha)\%$ CI, $\sigma_1 = \sigma_2$	$(\overline{x}_1 - \overline{x}_2) \pm t_{1-\alpha/2}\, s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}, \quad df = n_1 + n_2 - 2$	**pooled**
$100(1-\alpha)\%$ CI, may differ	$(\overline{x}_1 - \overline{x}_2) \pm t_{1-\alpha/2}\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}, \quad df \approx \min(n_1 - 1, n_2 - 1)$	**unpooled**

$s_p^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2)/(n_1 + n_2 - 2)$. **Default: unpooled ($H_0$ says nothing about σ)**

Example: girls, boys and sports

Girls and boys aged 7 to 11 are believed to spend the same average time each day playing sports. A study took a random sample of each; both populations are normal. Is there a difference in the mean time girls and boys spend playing sports each day? (OpenStax, Example 10.1)

girls: $n = 9,\ \overline{x} = 2$ **hours,** $s = 0.866$**;** **boys:** $n = 16,\ \overline{x} = 3.2$ **hours,** $s = 1.00$**;** $\alpha = 0.05$

(a) State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

(c) Find the P-value from the t-table. What is the decision?

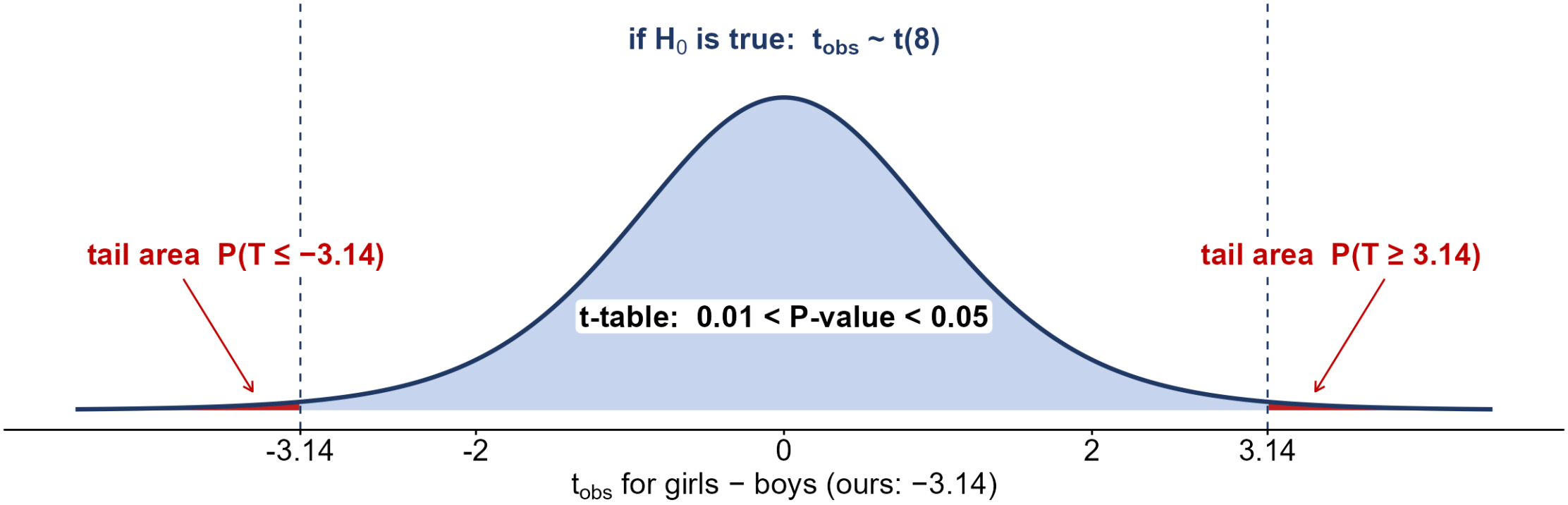
(a) $H_0: \mu_g - \mu_b = 0$ $H_A: \mu_g - \mu_b \neq 0$ **(two-sided)**

(b) $\text{SE} = \sqrt{0.866^2/9 + 1.00^2/16} = 0.3819;\ df = \min(8,15) = 8$

$t_{obs} = (2 - 3.2)/0.3819 = -3.14$

(c) $2.306 < 3.14 < 3.355:\ 0.01 <$ **P-value** $< 0.05:$ **reject** H_0

The sports P-value



$0.01 <$ **P-value** < 0.05 **(software: 0.0138): girls and boys differ in mean time playing sports**

Practice: who keeps workers longer?

A study asks whether Company A keeps its workers longer than Company B. Random samples of workers from each company give each worker's years with the company; both populations are normal. Does Company A keep its workers longer, on average? (OpenStax, Try It 10.2)

A: $n = 15,\ \overline{x} = 5$ **years,** $s = 1.2$**;** **B:** $n = 20,\ \overline{x} = 4.5$ **years,** $s = 0.8$**;** $\alpha = 0.05$

(a) State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

(c) Find the P-value from the t-table. What is the decision?

(a) $H_0: \mu_A - \mu_B = 0$ $H_A: \mu_A - \mu_B > 0$ **(upper-tailed)**

(b) $\text{SE} = \sqrt{1.2^2/15 + 0.8^2/20} = 0.3578;\ df = \min(14,19) = 14$

$t_{obs} = 0.5/0.3578 = 1.40$

(c) $1.345 < 1.40 < 1.523:\ 0.075 <$ **P-value** < 0.10**, above 0.05: do not reject** H_0

Example: pooling for girls and boys

Back to girls, boys and sports. The two sample SDs are close (0.866 and 1.00), so assuming $\sigma_g = \sigma_b$ is reasonable here. Redo the test with the pooled SD.

girls: $n = 9,\ s = 0.866$; **boys:** $n = 16,\ s = 1.00$; $\overline{x}_g - \overline{x}_b = -1.2$; $\alpha = 0.05$

(a) Find s_p and the pooled SE.

(b) Find t_{obs} and the df. Bracket the P-value.

(c) Compare with the unpooled test. What did pooling buy us?

(a) $s_p = \sqrt{(8(0.866^2) + 15(1.00^2))/23} = 0.9555;\ \text{SE} = 0.9555\sqrt{1/9 + 1/16} = 0.3981$

(b) $t_{obs} = -1.2/0.3981 = -3.01;\ df = 9 + 16 - 2 = 23$: $0.005 <$ **P-value** < 0.01

(c) unpooled: $df = 8,\ 0.01 <$ **P-value** < 0.05. **Pooling: more df, a smaller P-value (more power)**

Pooling: advantages and disadvantages

when it is reasonable

similar sample SDs (larger / smaller under about 2), similar sample sizes

the advantage

more df ($n_1 + n_2 - 2$) and an exact t distribution: more power if $\sigma_1 = \sigma_2$

the disadvantage

if $\sigma_1 \neq \sigma_2$ and the n's differ, the Type I error rate is not α

for example

$n_1 = 10, \sigma_1 = 3$ **vs** $n_2 = 40, \sigma_2 = 1$**: a 5% pooled test rejects a true H_0 26% of the time**